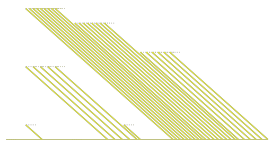


Chapitre 6 : La comparaison à une valeur théorique

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Table des matières

1 Avant de commencer	2
2 Théorème central limite	2
3 Comparaison d'une proportion a une valeur théorique	3
3.1 Exemple de comparaison	3
3.2 Exemple sur les pièces défectueuses	6
3.3 Exemple sur l'assemblée nationale	7
4 Comparaison d'une moyenne à une valeur théorique	9
4.1 distribution de la loi de Student	9
4.2 Exemple d'utilisation du test de Student	10
4.3 Test de Student apparié	11
4.4 Test de Wilcoxon apparié	14
5 Variance comparer à une valeur théorique	16
6 Test de Kolmogorov Smirnov	16



1 Avant de commencer

Ce chapitre reprends les codes du chapitre 6 sur **la comparaison à une valeur théorique** du livre **Langage R et Statistiques** : Initiation à l'analyse de données publié par les éditions ENI (2^{ème} édition).

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1      ✓ readr      2.2.0
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2     4.0.3      ✓ tibble     3.3.1
✓ lubridate  1.9.5      ✓ tidyr      1.3.2
✓ purrr       1.2.2

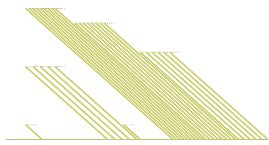
— Conflicts — tidyverse_conflicts() —
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

2 Théorème central limite

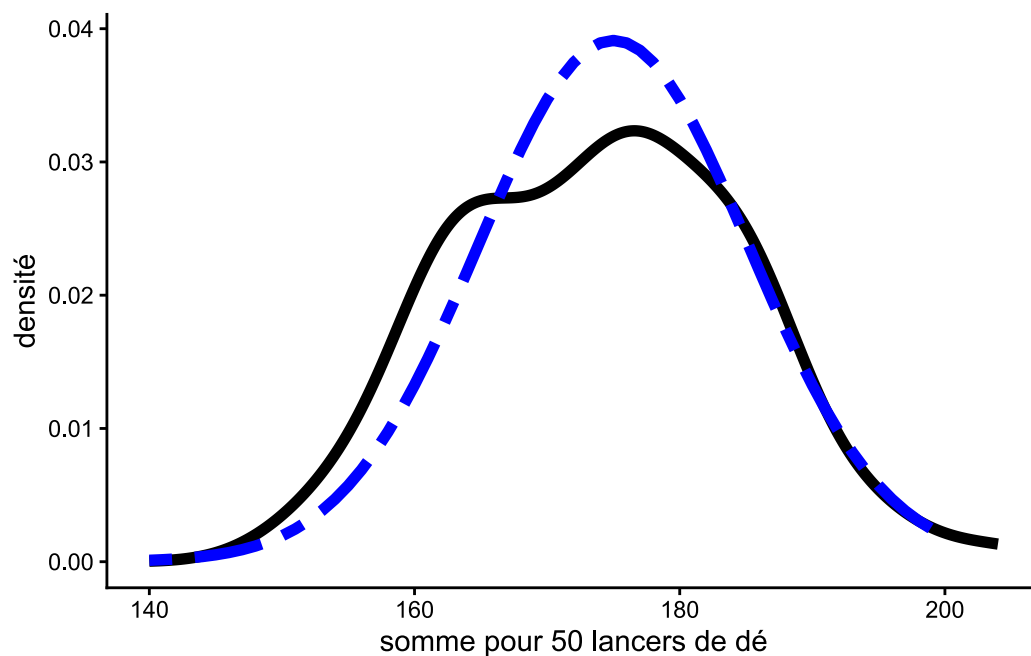
```
lancers_des <-
  purrr::map(
    .x = c(1:100),
    .f = ~ tibble(
      somme_des =
        runif(n = 50, min = 1, max = 6) |>
        round() |>
        sum()
    )
  ) |>
  bind_rows()

loi_normale <- tibble(
  x = seq(140, 200, 1),
  y = dnorm(x, mean = 175, sd = sqrt(104))
)

lancers_des |>
  ggplot() +
  aes(x = somme_des) +
```



```
geom_density(linewidth = 2) +  
geom_line(  
  aes(x = x, y = y),  
  data = loi_normale,  
  color = "blue",  
  linetype = "twodash",  
  linewidth = 2  
) +  
labs(x = "somme pour 50 lancers de dé", y = "densité") +  
theme_classic()
```

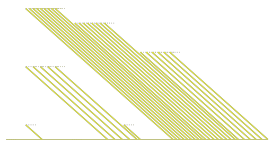


```
ggsave("img/06Rl01N.png", width = 14, height = 10, units = "cm")
```

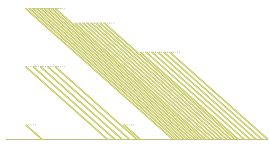
3 Comparaison d'une proportion a une valeur théorique

3.1 Exemple de comparaison

```
tibble(  
  z = seq(-3, 3, 0.1),  
  "densité" = dnorm(z, mean = 0, sd = 1)  
) |>  
ggplot() +  
aes(  
  z,  densité
```



```
x = z,  
y = `densité`  
) +  
geom_line(size = 1) +  
geom_vline(  
  xintercept = qnorm(0.025),  
  linewidth = 1,  
  linetype = "dashed",  
  color = "blue"  
) +  
geom_vline(  
  xintercept = qnorm(0.975),  
  linewidth = 1,  
  linetype = "dashed",  
  color = "blue"  
) +  
geom_vline(  
  xintercept = qnorm(0.05),  
  linewidth = 1,  
  linetype = "dotdash",  
  color = "green"  
) +  
geom_vline(  
  xintercept = qnorm(0.95),  
  linewidth = 1,  
  linetype = "dotted",  
  color = "orange"  
) +  
geom_vline(  
  xintercept = -2.3,  
  color = "red"  
) +  
geom_vline(  
  xintercept = -1.7,  
  color = "red"  
) +  
geom_vline(  
  xintercept = -0.6,  
  color = "red"  
) +  
geom_vline(  
  xintercept = 1.9,  
  color = "red"
```

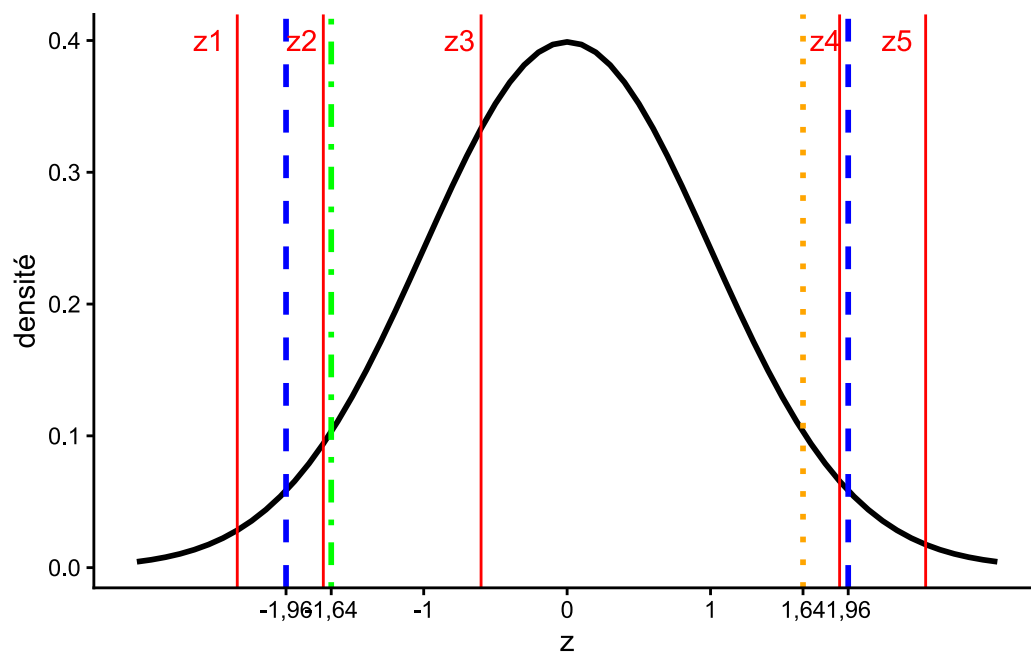


```

) +
geom_vline(
  xintercept = 2.5,
  color = "red"
) +
annotate("text", x = -2.5, y = 0.4, label = "z1", color = "red") +
annotate("text", x = -1.85, y = 0.4, label = "z2", color = "red") +
annotate("text", x = -0.75, y = 0.4, label = "z3", color = "red") +
annotate("text", x = 1.8, y = 0.4, label = "z4", color = "red") +
annotate("text", x = 2.3, y = 0.4, label = "z5", color = "red") +
scale_x_continuous(
  label = c("-1,96", "-1,64", "-1", "0", "1", "1,64", "1,96"),
  breaks = c(qnorm(0.025), qnorm(0.05), -1, 0, 1, qnorm(0.95), qnorm(0.975))
) +
theme_classic()

```

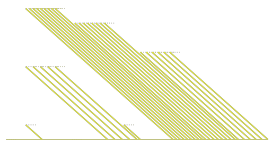
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` instead.



```

ggsave("img/06Rl02N.png", width = 14, height = 10, units = "cm")

```



3.2 Exemple sur les pièces défectueuses

comparaison a une proportion test

Les valeurs :

```
nbr_piece_defecteuse <- 22  
p_seuil <- 0.2  
n <- 100
```

statistique de test calculée à la main

```
(z <- (nbr_piece_defecteuse/n - p_seuil)/sqrt((p_seuil*(1-p_seuil))/n))
```

```
[1] 0.5
```

valeur seuil

```
qnorm(0.95)
```

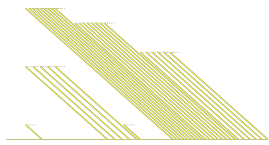
```
[1] 1.644854
```

utilisation de la fonction prop.test

```
prop.test(  
  x = nbr_piece_defecteuse,  
  n = n,  
  p = p_seuil,  
  alternative = "greater"  
)
```

1-sample proportions test with continuity correction

```
data:  nbr_piece_defecteuse out of n, null probability p_seuil  
X-squared = 0.14062, df = 1, p-value = 0.3538  
alternative hypothesis: true p is greater than 0.2  
95 percent confidence interval:  
 0.1554081 1.0000000  
sample estimates:  
  p  
0.22
```



utilisation de la fonction `binom.test()`

```
binom.test(  
  x = nbr_piece_defecteuse,  
  n = n,  
  p = p_seuil,  
  alternative = "greater"  
)
```

Exact binomial test

```
data:  nbr_piece_defecteuse and n  
number of successes = 22, number of trials = 100, p-value = 0.346  
alternative hypothesis: true probability of success is greater than 0.2  
95 percent confidence interval:  
 0.153889 1.000000  
sample estimates:  
probability of success  
      0.22
```

3.3 Exemple sur l'assemblée nationale

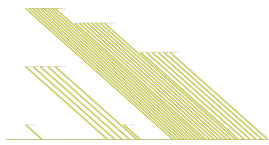
```
library(rvest)
```

Attachement du package : 'rvest'

L'objet suivant est masqué depuis 'package:readr':

guess_encoding

```
html  <-  read_html("https://www2.assemblee-nationale.fr/deputes/liste/cat-  
sociopro/(vue)/tableau")  
  
(prop_ass_nat <- html |>  
  html_elements("table") |>  
  html_table() |>  
  bind_rows() |>  
  count(Civilité))
```



```
# A tibble: 2 × 2
  Civilité      n
  <chr>    <int>
1 M.        367
2 Mme       207
```

calcul de la proportion d'hommes à l'Assemblée

```
(p_ass <- prop_ass_nat[[1,2]] / sum(prop_ass_nat$n))
```

```
[1] 0.6393728
```

population française 1er janvier 2026 : <https://www.insee.fr/fr/statistiques/2381474>

```
(p_pop <- 33551000 / (35531000 + 33551000))
```

```
[1] 0.4856692
```

calcul de la statistique de test

```
(z <- (p_ass - p_pop)/sqrt((p_pop*(1-p_pop))/sum(prop_ass_nat$n)))
```

```
[1] 7.367981
```

comparaison aux valeurs de seuil

```
z < qnorm(0.025)
```

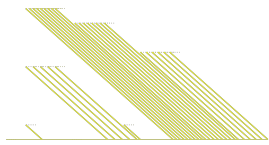
```
[1] FALSE
```

```
z > qnorm(0.975)
```

```
[1] TRUE
```

utilisation de la fonction `prop.test()`

```
prop.test(
  x = p_ass,
  n = sum(prop_ass_nat$n),
```



```
p = p_pop,  
alternative = "two.sided"  
)
```

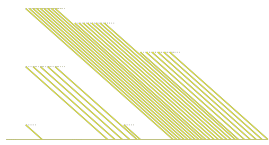
1-sample proportions test with continuity correction

```
data:  p_ass out of sum(prop_ass_nat$n), null probability p_pop  
X-squared = 537.59, df = 1, p-value < 2.2e-16  
alternative hypothesis: true p is not equal to 0.4856692  
95 percent confidence interval:  
 8.222931e-06 1.020808e-02  
sample estimates:  
      p  
0.00111389
```

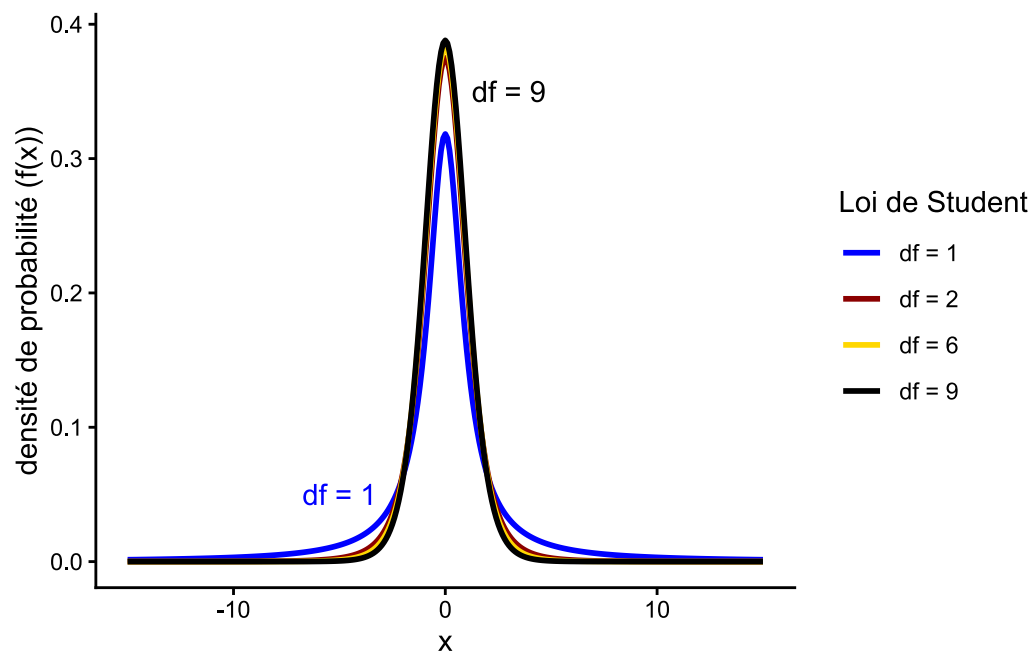
4 Comparaison d'une moyenne à une valeur théorique

4.1 distribution de la loi de Student

```
# choix de l'intervalle de représentation  
intervalle <- seq(-15, 15, 0.1)  
  
# loi de Student de degré de liberté égale à 1, 2, 6 et 9  
tibble(  
  loi_student =  
    rep(  
      c(  
        "df = 1",  
        "df = 2",  
        "df = 6",  
        "df = 9"  
      ),  
      each = length(intervalle)  
    ),  
  x = rep(intervalle, 4),  
  densite =  
    c(  
      dt(intervalle, df = 1),  
      dt(intervalle, df = 4),  
      dt(intervalle, df = 6),  
      dt(intervalle, df = 9)  
    )  
)
```



```
)  
) |>  
ggplot() +  
aes(x = x, y = densite, color = loi_student) +  
geom_line(linewidth = 1) +  
annotate("text", x = -5, y = 0.05, label = "df = 1", color = "blue") +  
annotate("text", x = 3, y = 0.35, label = "df = 9", color = "black") +  
theme_classic() +  
scale_color_manual(values = c("blue", "dark red", "gold", "black")) +  
ylab("densité de probabilité (f(x))") +  
labs(color = "Loi de Student")
```



```
ggsave("img/06Rl06N.png", width = 14, height = 10, units = "cm")
```

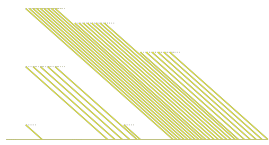
4.2 Exemple d'utilisation du test de Student

calcul_moyenne

```
mean(women$weight)
```

```
[1] 136.7333
```

vérification de la normalité de la variable



```
shapiro.test(women$weight)
```

Shapiro-Wilk normality test

```
data:  women$weight  
W = 0.96036, p-value = 0.6986
```

test de Student, la masse moyenne de l'échantillon est-elle égale à celle de la population ?

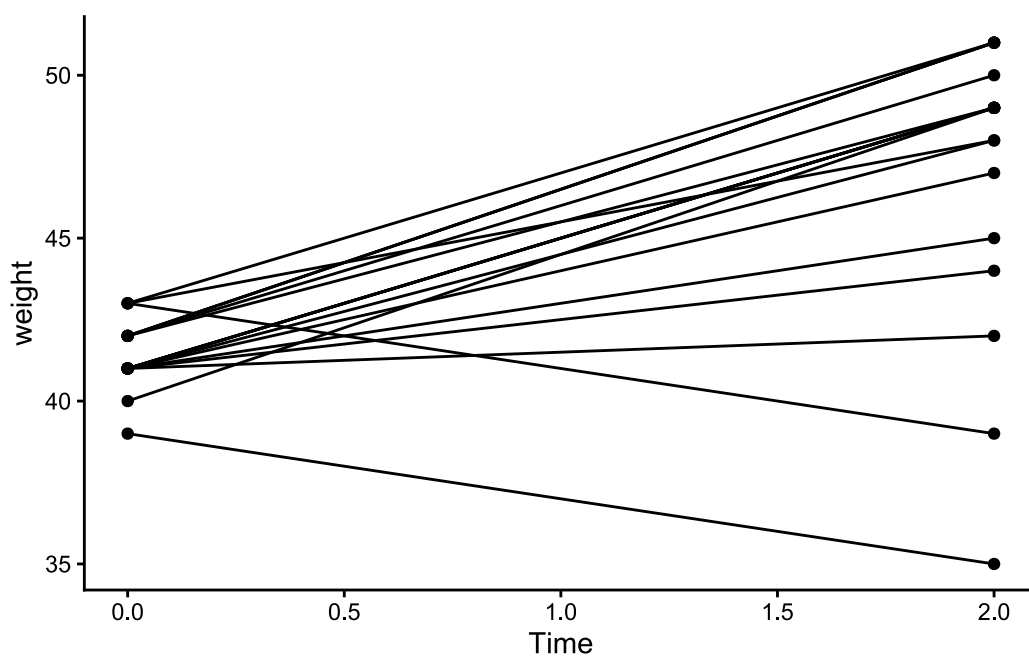
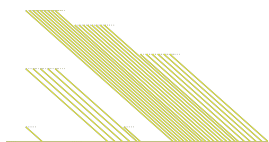
```
t.test(  
  x = women$weight,  
  mu = 79.3*2.2  
)
```

One Sample t-test

```
data:  women$weight  
t = -9.4276, df = 14, p-value = 1.929e-07  
alternative hypothesis: true mean is not equal to 174.46  
95 percent confidence interval:  
 128.1504 145.3162  
sample estimates:  
mean of x  
136.7333
```

4.3 Test de Student apparié

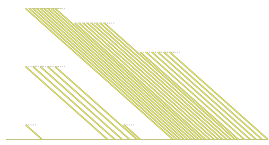
```
ChickWeight |>  
  filter(  
    Diet == 1,  
    Time %in% c(0, 2)  
  ) |>  
  ggplot() +  
  aes(x = Time, y = weight, group = Chick) +  
  geom_point() +  
  geom_line() +  
  theme_classic()
```



création données

```
(poulet_croissance <- ChickWeight |>
  filter(
    Diet == 1,
    Time %in% c(0, 2)
  ) |>
  pivot_wider(
    id_cols = !Diet,
    names_from = Time,
    values_from = weight
  ) |>
  mutate(difference = `2` - `0`))
```

```
# A tibble: 20 × 4
  Chick `0` `2` difference
  <ord> <dbl> <dbl>     <dbl>
1 1 42 51 9
2 2 40 49 9
3 3 43 39 -4
4 4 42 49 7
5 5 41 42 1
6 6 41 49 8
7 7 41 49 8
```



8 8	42	50	8
9 9	42	51	9
10 10	41	44	3
11 11	43	51	8
12 12	41	49	8
13 13	41	48	7
14 14	41	49	8
15 15	41	49	8
16 16	41	45	4
17 17	42	51	9
18 18	39	35	-4
19 19	43	48	5
20 20	41	47	6

```
# test de normalité des données
shapiro.test(poulet_croissance$difference)
```

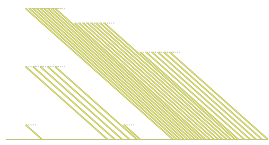
Shapiro-Wilk normality test

```
data: poulet_croissance$difference
W = 0.7411, p-value = 0.0001293
```

```
# test de Student apparié à ne pas faire normalement
t.test(poulet_croissance`0`, poulet_croissance`2`, paired = TRUE)
```

Paired t-test

```
data: poulet_croissance`0` and poulet_croissance`2`
t = -6.5346, df = 19, p-value = 2.94e-06
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -7.72375 -3.97625
sample estimates:
mean difference
 -5.85
```



4.4 Test de Wilcoxon apparié

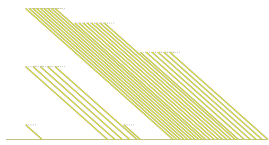
```
# ajout des rangs signés
(test_wilcoxon <- poulet_croissance |>
  filter(difference != 0) |>
  arrange(abs(difference)) |>
  rowid_to_column(var = "rang") |>
  mutate(rang_signe = rang*sign(difference)))
```

```
# A tibble: 20 × 6
  rang Chick `0` `2` difference rang_signe
  <int> <ord> <dbl> <dbl>      <dbl>      <dbl>
1     1  1 5    41  42         1         1
2     2  2 10   41  44         3         2
3     3  3 3    43  39        -4        -3
4     4  4 16   41  45         4         4
5     5  5 18   39  35        -4        -5
6     6  6 19   43  48         5         6
7     7  7 20   41  47         6         7
8     8  8 4    42  49         7         8
9     9  9 13   41  48         7         9
10    10 10 6   41  49         8        10
11    11 11 7   41  49         8        11
12    12 12 8   42  50         8        12
13    13 13 11  43  51         8        13
14    14 14 12   41  49         8        14
15    15 15 14   41  49         8        15
16    16 16 15   41  49         8        16
17    17 17 1    42  51         9        17
18    18 18 2    40  49         9        18
19    19 19 9    42  51         9        19
20    20 20 17   42  51         9        20
```

```
# calcul des paramètres de la statistique de test
(n <- nrow(test_wilcoxon))
```

```
[1] 20
```

```
(somme_rang_positif <- test_wilcoxon |>
  filter(rang_signe > 0) |>
  summarise(sum(rang_signe)))
```



```
# A tibble: 1 × 1
  `sum(rang_signe)`
      <dbl>
1             202
```

```
(somme_rang_negatif <- test_wilcoxon |>
  filter(rang_signe < 0) |>
  summarise(sum(rang_signe)))
```

```
# A tibble: 1 × 1
  `sum(rang_signe)`
      <dbl>
1             -8
```

```
(T_calcule <- min(abs(somme_rang_positif), abs(somme_rang_negatif)))
```

```
[1] 8
```

```
# calcul de la statistique de test
(z <- (T_calcule - (n*(n+1))/4)/sqrt((n*(n+1)*(2*n+1))/24))
```

```
[1] -3.621269
```

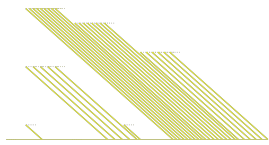
```
# conclusion
z < qnorm(0.05)
```

```
[1] TRUE
```

```
# test de wilcoxon apparié
wilcox.test(poulet_croissance$`0`, poulet_croissance$`2`, paired = TRUE)
```

Wilcoxon signed rank exact test

data: poulet_croissance\$`0` and poulet_croissance\$`2`
V = 8, p-value = 4.578e-05
alternative hypothesis: true location shift is not equal to 0



5 Variance comparer à une valeur théorique

calcul de la variance

```
var(women$weight)
```

```
[1] 240.2095
```

comparaison d'une variance à une valeur de référence

```
# données
n <- nrow(PlantGrowth)
var_echantillon <- var(PlantGrowth$weight)
var_theo <- 0.35

# calcul de la statistique de test
((n-1)*var_echantillon/var_theo)
```

```
[1] 40.73837
```

```
# récupération de la valeur seuil pour alpha = 0,05
qchisq(p = 1-0.05, df = n-1)
```

```
[1] 42.55697
```

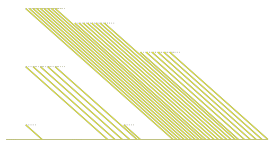
6 Test de Kolmogorov Smirnov

comparaison avec le test de kuiper

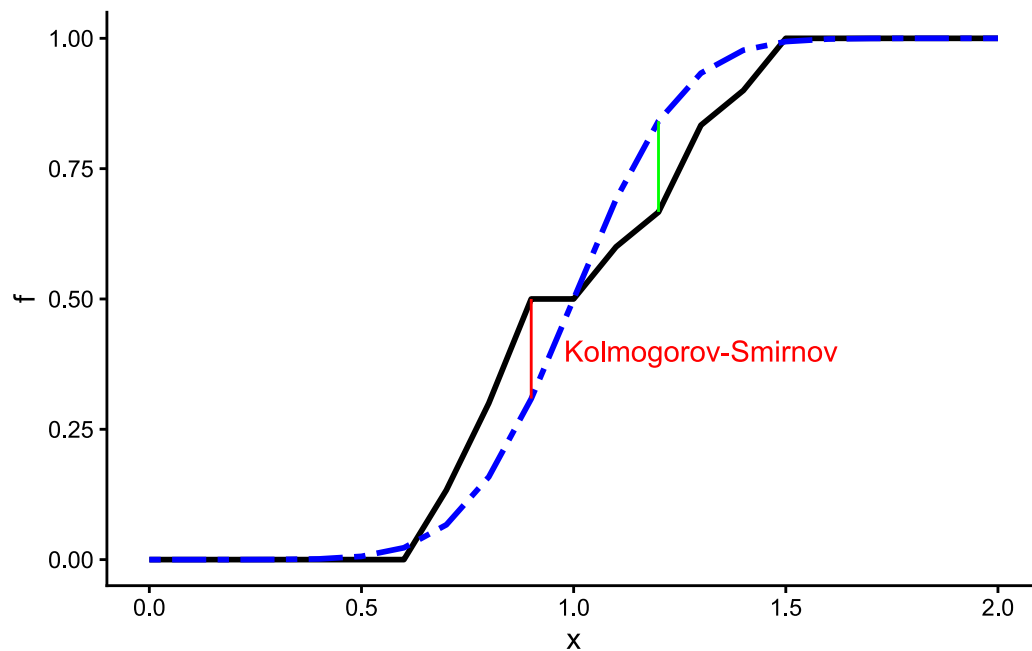
```
set.seed(1903)

echantillon <- 0.5 + 1.1 * runif(30)
fonction_repartition <- ecdf(echantillon)

tibble(
  x = seq(0, 2, 0.1),
  f = fonction_repartition(x),
  densite_normale = pnorm(x, mean = 1, sd = 0.2)
) |>
ggplot() +
  aes(x = x, y = f) +
  geom_line(linewidth = 1) +
```



```
geom_line(  
  aes( y = densite_normale),  
  color = "blue",  
  linewidth = 1,  
  linetype = "twodash"  
) +  
  annotate("segment", x = 1.2, xend = 1.2, y = 0.667, yend = 0.841, color =  
"green") +  
  annotate("segment", x = 0.9, xend = 0.9, y = 0.309, yend = 0.5, color = "red") +  
  annotate("text", x = 1.3, y = 0.4, color = "red", label = "Kolmogorov-Smirnov") +  
  theme_classic()
```

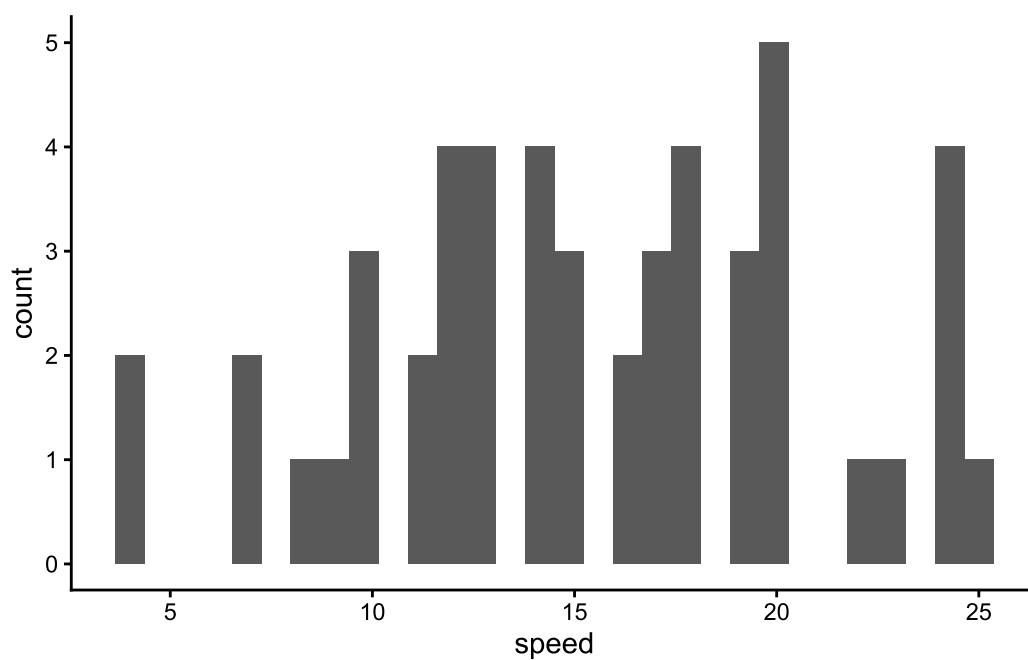
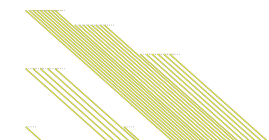


```
ggsave("img/06Rl18N.png", width = 14, height = 10, units = "cm")
```

test kolmogorov-smirnov

```
ggplot(cars) +  
  aes(x = speed) +  
  geom_histogram() +  
  theme_classic()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
ks.test(cars$speed, "pgamma", shape = 14)
```

Warning in ks.test.default(cars\$speed, "pgamma", shape = 14): ties should not be present for the one-sample Kolmogorov-Smirnov test

Asymptotic one-sample Kolmogorov-Smirnov test

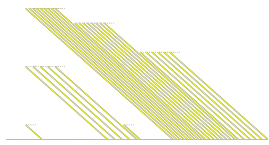
```
data: cars$speed
D = 0.23913, p-value = 0.006572
alternative hypothesis: two-sided
```

```
ks.test(cars$speed, "pgamma", shape = 15)
```

Warning in ks.test.default(cars\$speed, "pgamma", shape = 15): ties should not be present for the one-sample Kolmogorov-Smirnov test

Asymptotic one-sample Kolmogorov-Smirnov test

```
data: cars$speed
```



```
D = 0.17192, p-value = 0.1041
alternative hypothesis: two-sided
```

```
ks.test(cars$speed, "pnorm", mean = 15, sd = 2)
```

```
Warning in ks.test.default(cars$speed, "pnorm", mean = 15, sd = 2): ties should
not be present for the one-sample Kolmogorov-Smirnov test
```

Asymptotic one-sample Kolmogorov-Smirnov test

```
data: cars$speed
D = 0.31319, p-value = 0.0001099
alternative hypothesis: two-sided
```

```
ks.test(cars$speed, "pnorm", mean = 15, sd = 4)
```

```
Warning in ks.test.default(cars$speed, "pnorm", mean = 15, sd = 4): ties should
not be present for the one-sample Kolmogorov-Smirnov test
```

Asymptotic one-sample Kolmogorov-Smirnov test

```
data: cars$speed
D = 0.15337, p-value = 0.1901
alternative hypothesis: two-sided
```

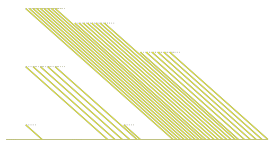
test kolmogorov-smirnov comparaison

```
ks.test(cars$speed, cars$dist)
```

Exact two-sample Kolmogorov-Smirnov test

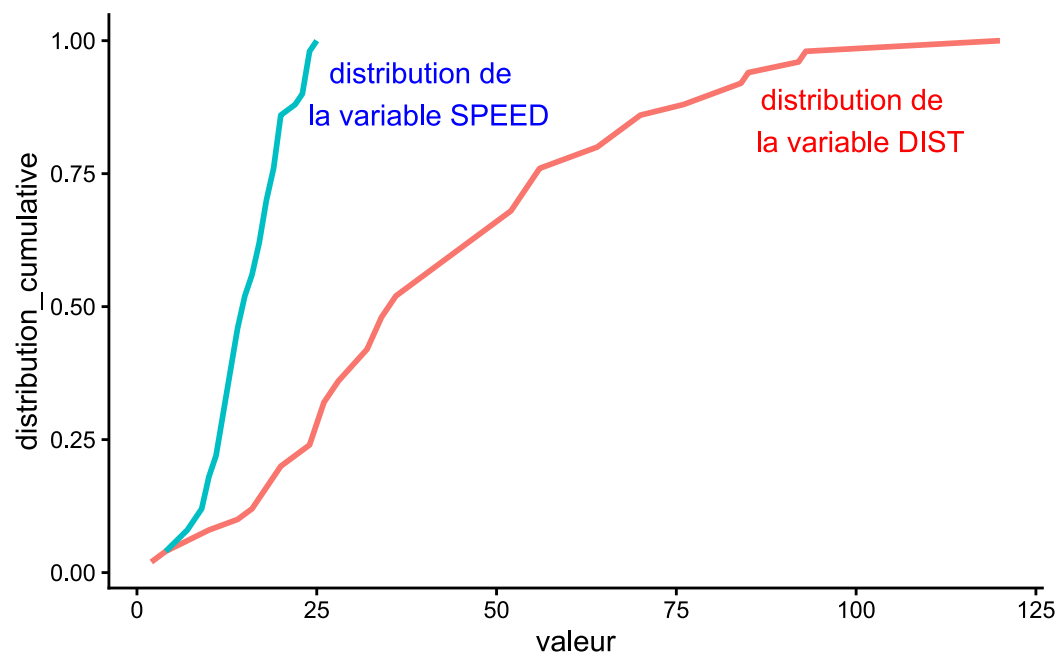
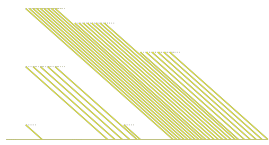
```
data: cars$speed and cars$dist
D = 0.76, p-value = 5.995e-15
alternative hypothesis: two-sided
```

graphique



```
fonction_repartition_speed <- ecdf(cars$speed)
fonction_repartition_dist <- ecdf(cars$dist)

bind_rows(
  cars |>
    mutate(
      distribution_speed = fonction_repartition_speed(speed),
    ) |>
    select(-dist) |>
    pivot_longer(
      !speed,
      names_to = "type_distribution",
      values_to = "distribution_cumulative"
    ) |>
    rename("valeur" = "speed"),
  cars |>
    mutate(
      distribution_dist = fonction_repartition_dist(dist),
    ) |>
    select(-speed) |>
    pivot_longer(
      !dist,
      names_to = "type_distribution",
      values_to = "distribution_cumulative"
    ) |>
    rename("valeur" = "dist")
) |>
ggplot() +
  aes(x = valeur, y = distribution_cumulative, color = type_distribution) +
  geom_line(linewidth = 1) +
  annotate("text", x = 100, y = 0.85, color = "red", label = "distribution de \n la variable DIST") +
  annotate("text", x = 40, y = 0.9, color = "blue", label = "distribution de \n la variable SPEED") +
  theme_classic() +
  theme(legend.position = "none")
```



```
ggsave("img/06Rl25N.png", width= 14, height = 10, units = "cm")
```